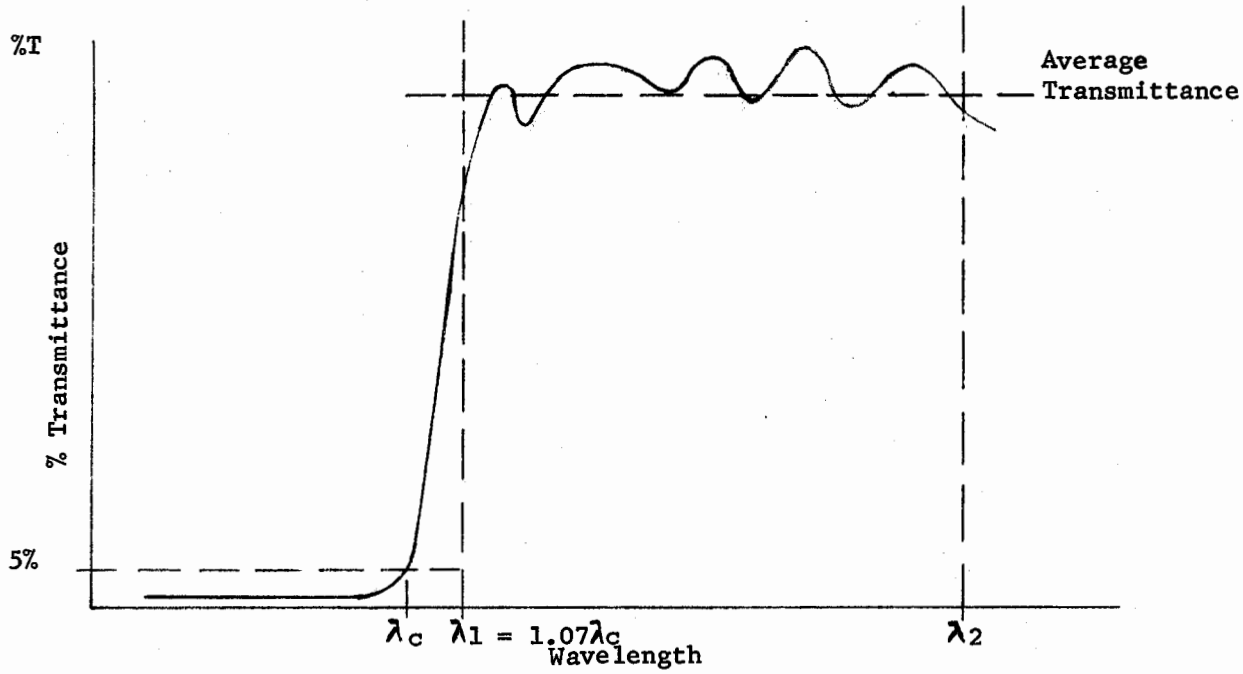
MILITARY SPECIFICATION SHEET

COATING, LONG WAVELENGTH PASS FILTER, ORDINARY REQUIREMENTS FOR

The complete requirements for procuring the long wavelength pass filters described herein shall consist of this document and the latest issue of MIL-F-48616.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The following figure illustrates the terms used for describing long wavelength pass filters.



NOTES FOR FIGURE:

1. The cut-on wavelength (λ_c) is the wavelength at which the absolute transmittance is 5%.

2. Average transmittance is calcuated between λ_1, & λ_2 where $\lambda_1 = 1.07\ \lambda_c$ and λ_2 is specified by the user.

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Long Wavelength Pass Filter Requirements

TABLE I

Cut-on Wavelength	Longest Wavelength of Transmitting Region	Minimum Average Transmittance	Dash No.
0.7 to 1.1 um	0.75 to 1.1 um	85%	1
	1.1 to 2.2 um	80%	2
	2.2 to 3.5 um	75%	3
1.1 to 1.8 um	1.18 to 1.8 um	85%	4
	1.8 to 3.0 um	80%	5
	3.0 to 5.0 um	75%	6
1.8 to 6.0 um	1.93 to 3.0 um	80%	7
	3.0 to 5.0 um	75%	8
	5.0 to 8.0 um	70%	9
6.0 to 14.0 um	6.45 to 12.0 um	80%	10
	12.0 to 14.0 um	75%	11
	14.0 to 20.0 um	65%	12
14.0 to 25.0 um	15.0 to 22.0 um	70%	13
	22.0 to 30.0 um	60%	14
	30.0 to 50.0 um	50%	15
25.0 to 45.0 um	26.9 to 30.0 um	60%	16
	30.0 to 50.0 um	50%	17

NOTES FOR REQUIREMENTS TABLE:

1. The 5% absolute transmittance point λ_c shall be located to within $\pm$ 2% of the specified wavelength.

2. Transmittance shall be equal to or less than 0.1% absoulte from 0.3 um to 0.93 λ_c .

3. The values in this table apply at an angle of incidence ranging from 0^o to 5^o and at ambient temperature. Measurements will be made at an angle of incidence within this range (See 6.4b and 6.4c).

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4. Spectral resolution to verify the above requirements in the cut-on and transmitting regions shall be 0.5% or less of the cut-on wavelength. Resolution shall be 10% or less of the cut on wavelength in the attenuation region. Scan speed and time constant shall be set such that the spectral features of interest can be resolved within the accuracy of 4.4.2.3.2.2 of MIL-F-48616.

Ordering Data:

At a minimum, the cut-on wavelength (See 6.2m) (λ_c) and the longest wavelength of the transmitting region (See 6.2m) (λ_c) and the dash number (See 6.2b) shall be specified by the procuring activity.

Preparing actitity:

Army - AR

Project No. 6650-0091-6

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